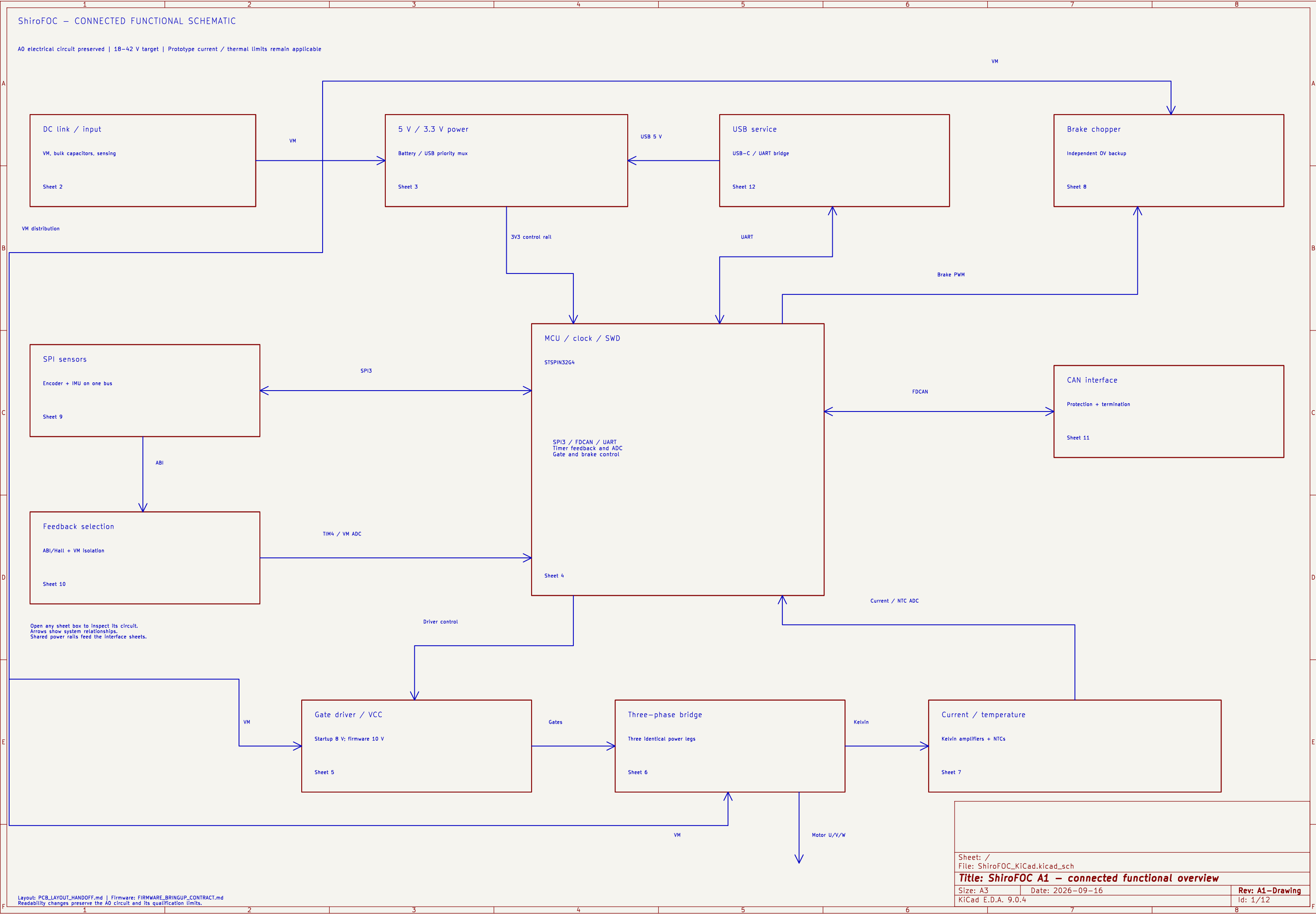




A

B

C

D

E

F

A

B

C

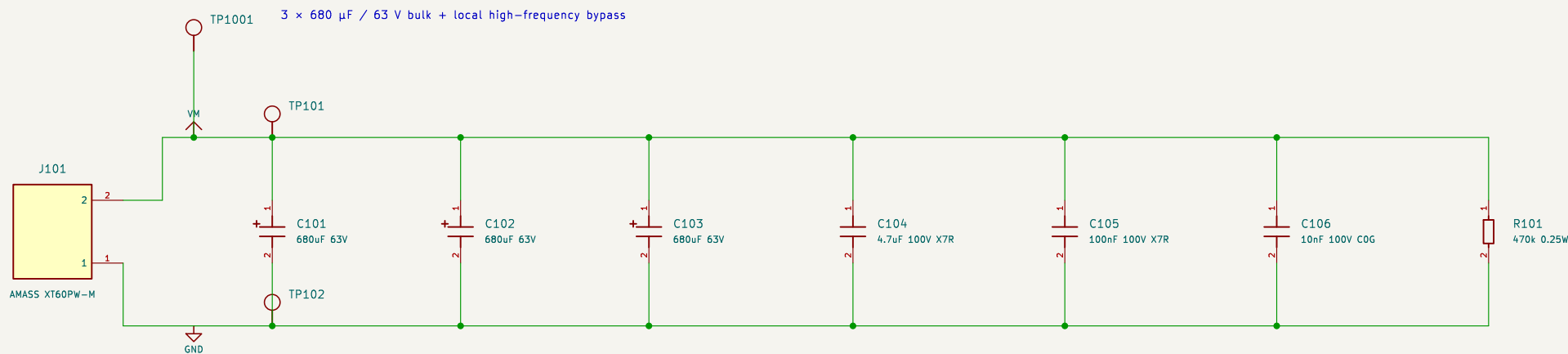
D

E

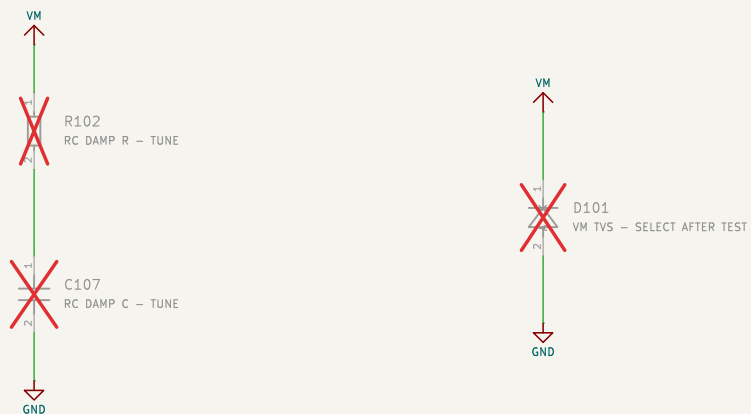
F

BATTERY / DC LINK

18–42 V operating bus • keyed XT60 • external fuse / precharge required

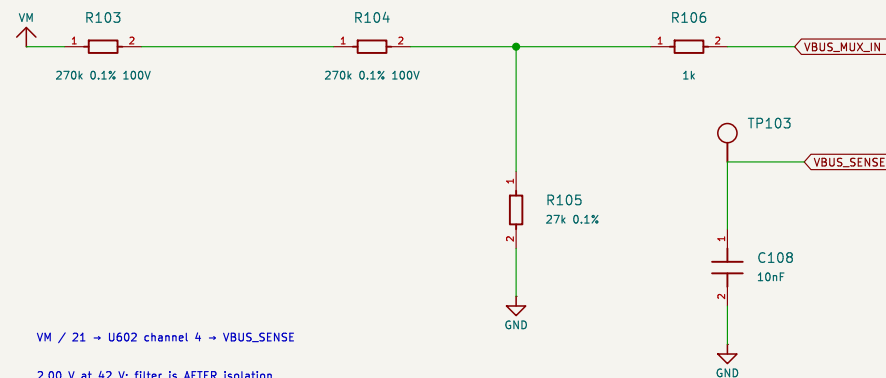


OPTIONAL INPUT DAMPING



DNP: choose clamp and damping only after cable / hot-plug tests.

BUS DIVIDER / ISOLATED ADC FILTER



VM / 21 → U602 channel 4 → VBUS_SENSE

2.00 V at 42 V; filter is AFTER isolation.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /DC link / input/

File: 01_Battery_DC_Link.kicad_sch

Title: Battery input, DC link and bus measurement

Size: A4 Date: 2026-09-16

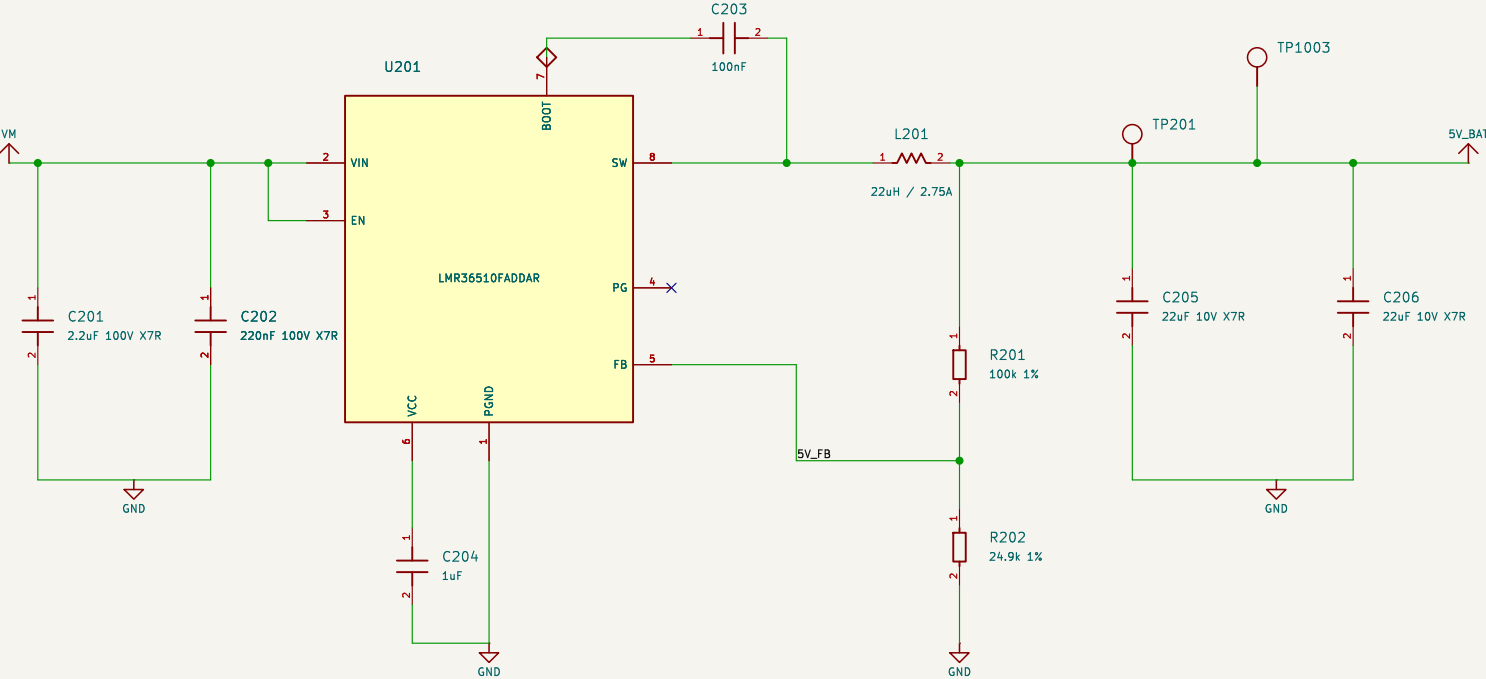
KiCad E.D.A. 9.0.4

Rev: A1-Drawing

Id: 2/12

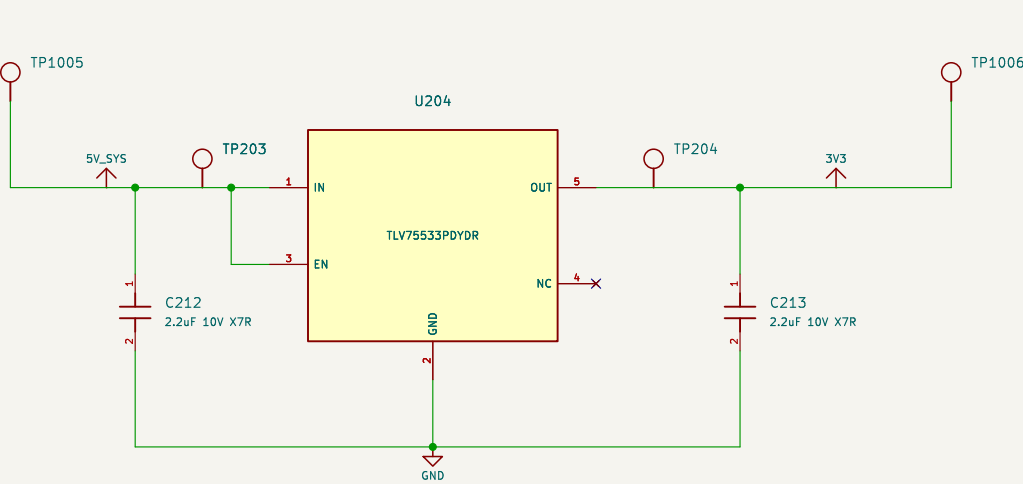
AUXILIARY POWER – USB HAS PRIORITY

BATTERY → 5 V BUCK



FB loop: 100 kΩ / 24.9 kΩ → approximately 5.02 V

5 V SYSTEM → 3.3 V

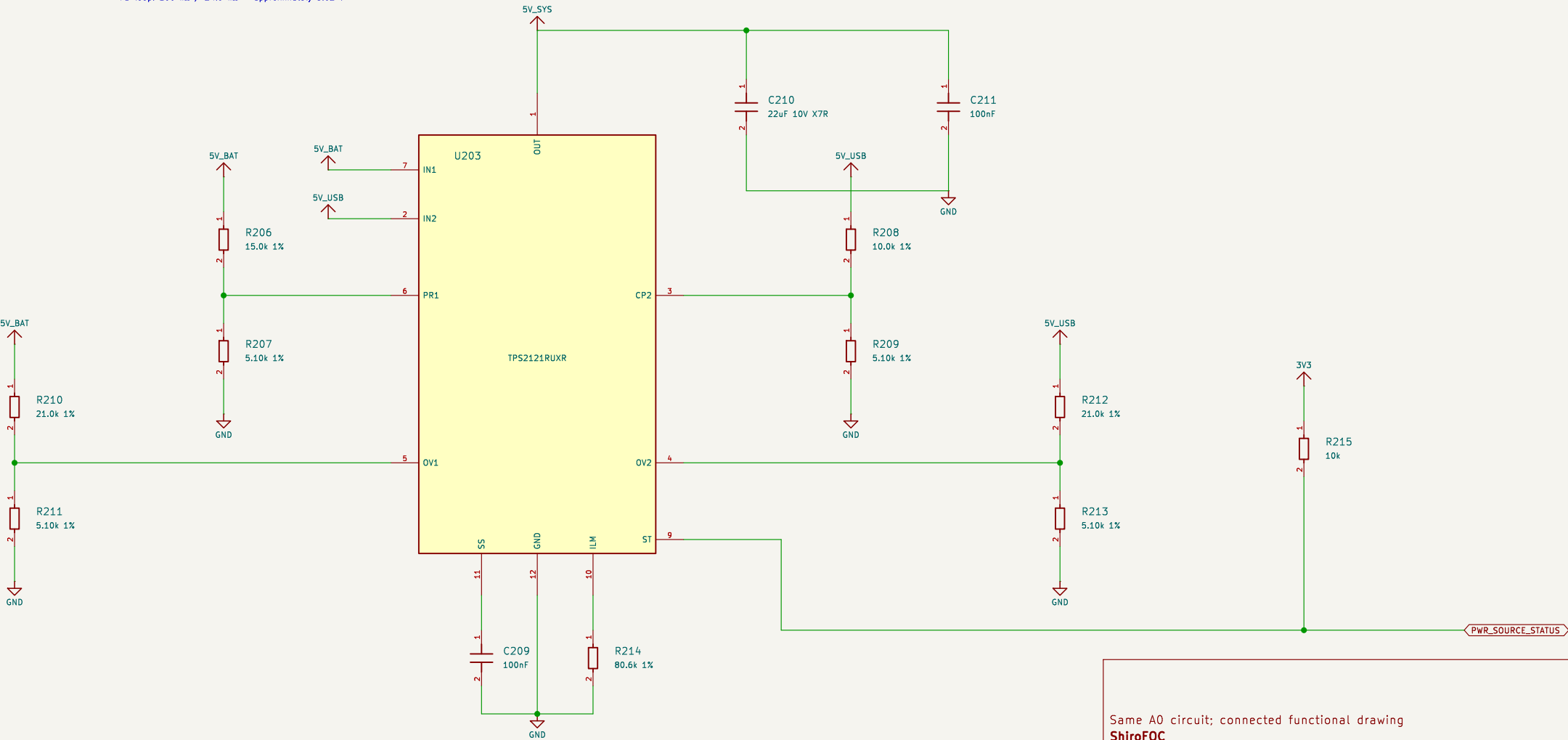


Debugger VTREF is sense-only; never inject 3V3.

U204 thermal pad / pin 2 returns to GND.

USB-PRIORITY POWER MUX

5V_USB arrives from the USB service sheet.



At 5 V: PR1 = 1.269 V; CP2 = 1.689 V.

USB wins the priority comparison. OV ≈ 5.43 V.

80.6 kΩ sets approximately 1.5 A current limit.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /5 V / 3.3 V power/

File: 02_Auxiliary_Power.kicad_sch

Title: Auxiliary supplies / USB priority

Size: A3

Date: 2026-09-16

Rev: A1-Drawing

KiCad E.D.A. 9.0.4

Id: 3/12

CONTROLLER / CLOCK / DEBUG

A

B

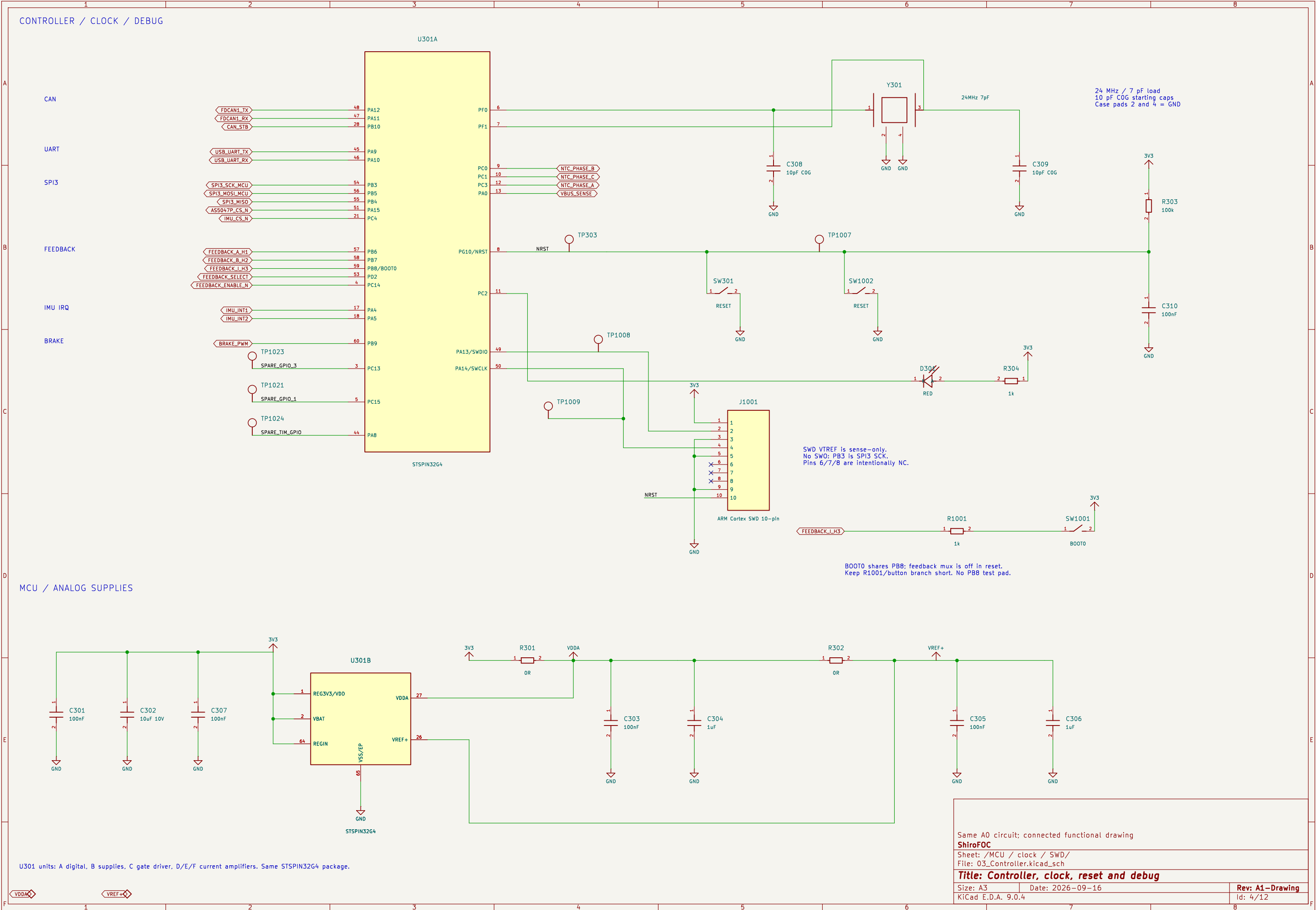
C

D

MCU / ANALOG SUPPLIES

E

F



Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /MCU / clock / SWD/

File: 03_Controller.kicad_sch

Title: Controller, clock, reset and debug

Size: A3

Date: 2026-09-16

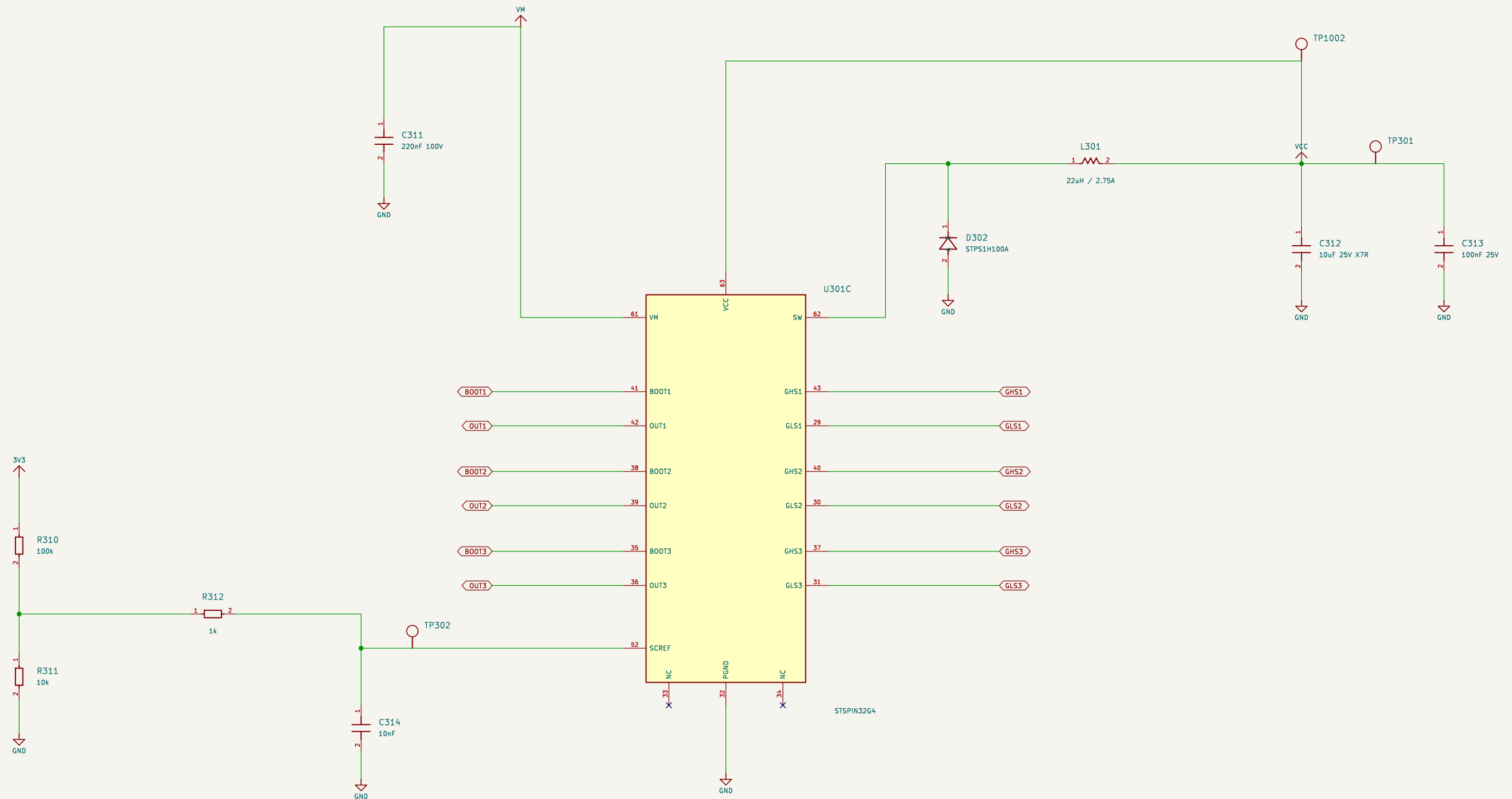
Rev: A1-Drawing

KiCad E.D.A. 9.0.4

Id: 4/12

GATE DRIVER / VCC CONVERTER

VDS FAULT REFERENCE



SCREF is a VDS fault threshold (~0.30 V), not a precision phase-current limit. Configure COMP1/2/4 and TIM1 break before enabling PWM.

VCC starts at 8 V. Reapply the 10 V setting after each VM return.
REGIN and REG3V3/VDD use external 3V3 (unit B). Gate supply stays off with USB alone.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /Gate driver / VCC/

File: 04_Gate_Driver.kicad_sch

Title: STSPIN gate drive and 10 V converter

Size: A3

Date: 2026-09-16

Rev: A1-Drawing

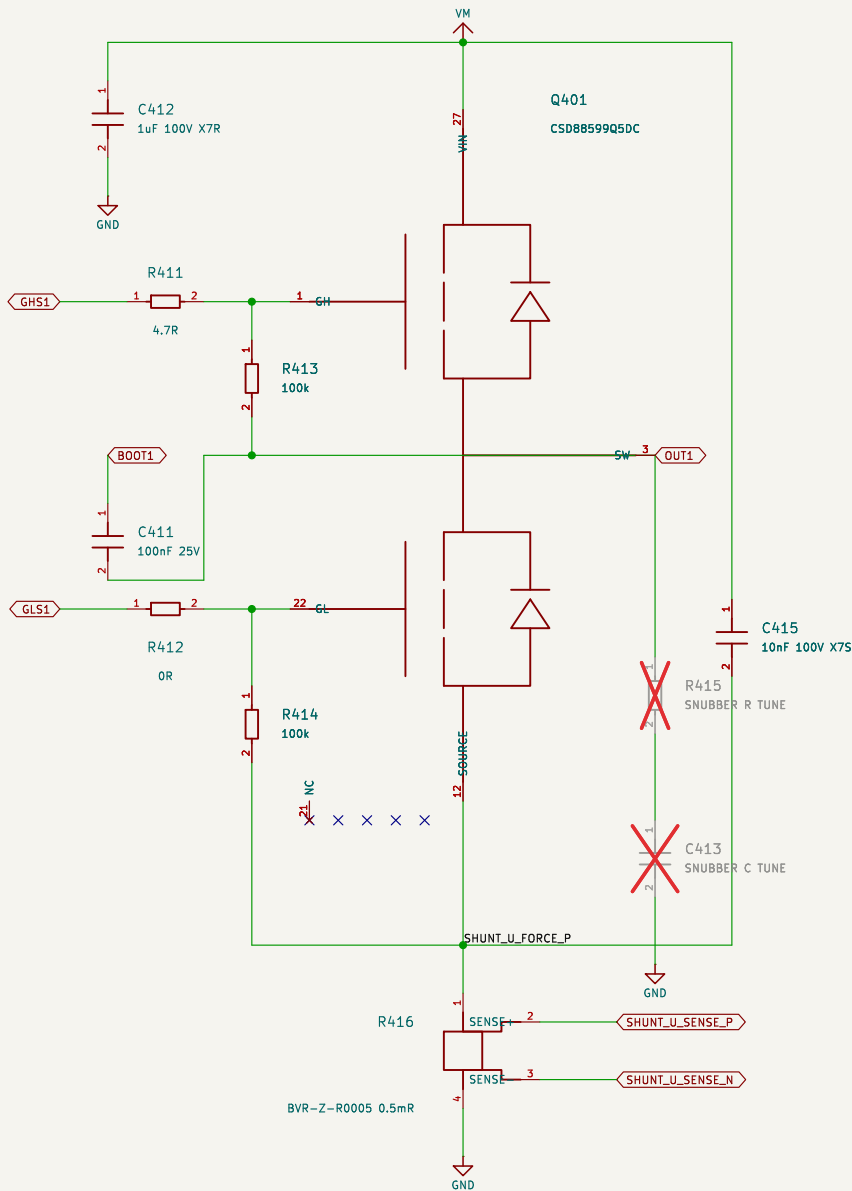
KiCad E.D.A. 9.0.4

Id: 5/12

THREE IDENTICAL PHASE POWER LOOPS

Gate resistors at MOSFET pins. The 10 nF capacitor returns above the shunt; the 1 uF capacitor returns to GND.

PHASE U

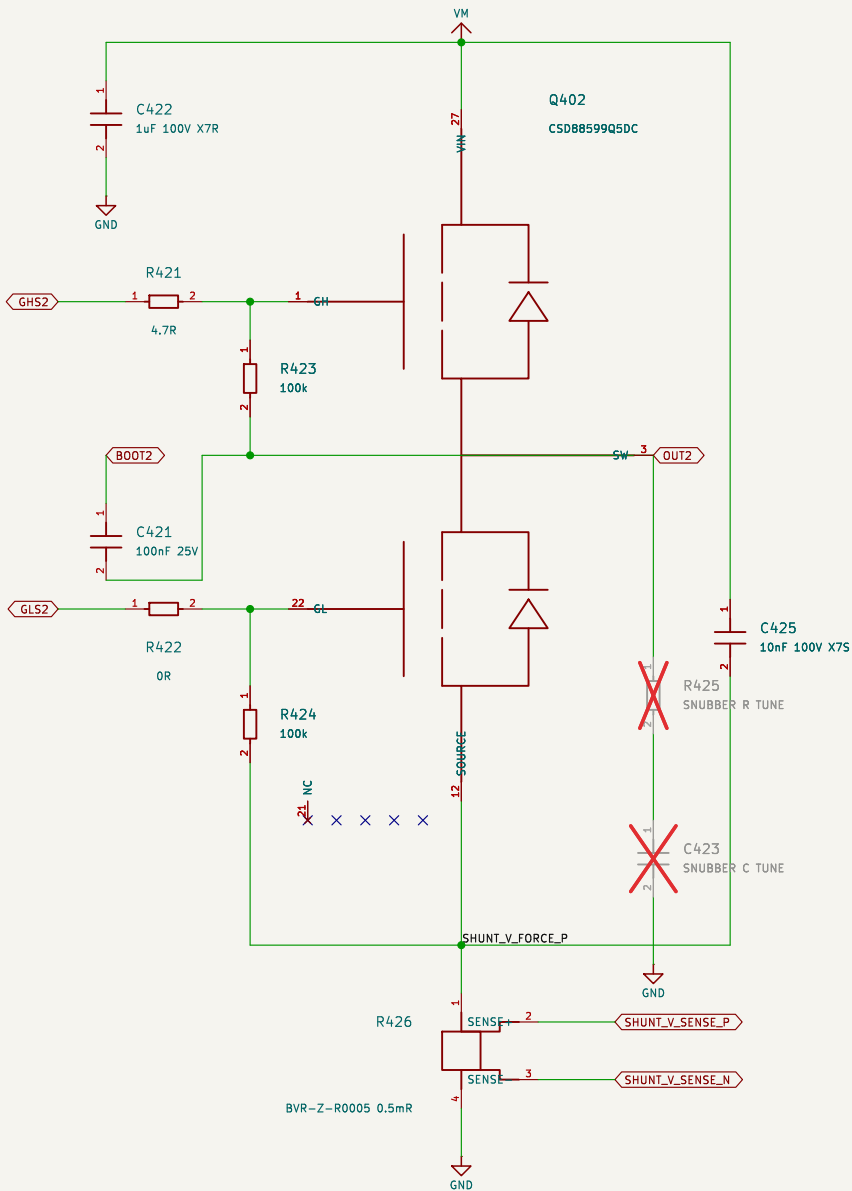


Package pins: VIN 27; GH 1; GL 22.
SW/SH: 2, 3-11; source: 12-20.
NC: 21, 23-26. Stacked pins are retained.

Kelvin pads 2/3 go only to the amplifier.
Force-current copper uses pads 1/4.
Snubber R/C are DNP tuning positions.

Layout order: VM ceramic -> high-side FET -> low-side FET -> local source return,
Keep the bootstrap and gate loops short; do not route motor current in Kelvin sense traces.

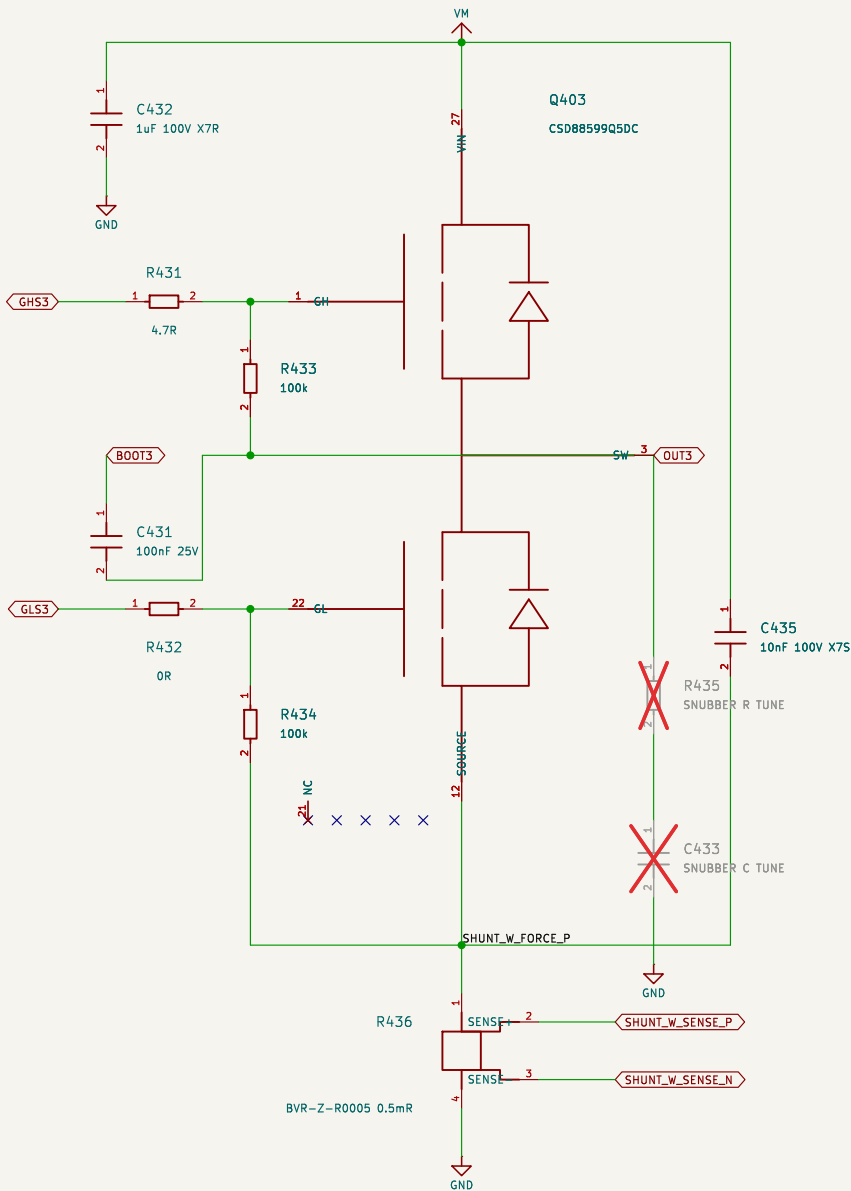
PHASE V



Package pins: VIN 27; GH 1; GL 22.
SW/SH: 2, 3-11; source: 12-20.
NC: 21, 23-26. Stacked pins are retained.

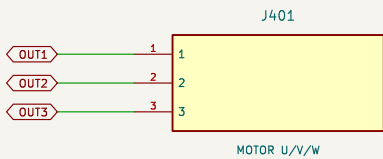
Kelvin pads 2/3 go only to the amplifier.
Force-current copper uses pads 1/4.
Snubber R/C are DNP tuning positions.

PHASE W



Package pins: VIN 27; GH 1; GL 22.
SW/SH: 2, 3-11; source: 12-20.
NC: 21, 23-26. Stacked pins are retained.

Kelvin pads 2/3 go only to the amplifier.
Force-current copper uses pads 1/4.
Snubber R/C are DNP tuning positions.



Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /Three-phase bridge/

File: 04_Inverter.kicad_sch

Title: Three-phase inverter / gate loops / Kelvin shunts

Size: A3

Date: 2026-09-16

Rev: A1-Drawing

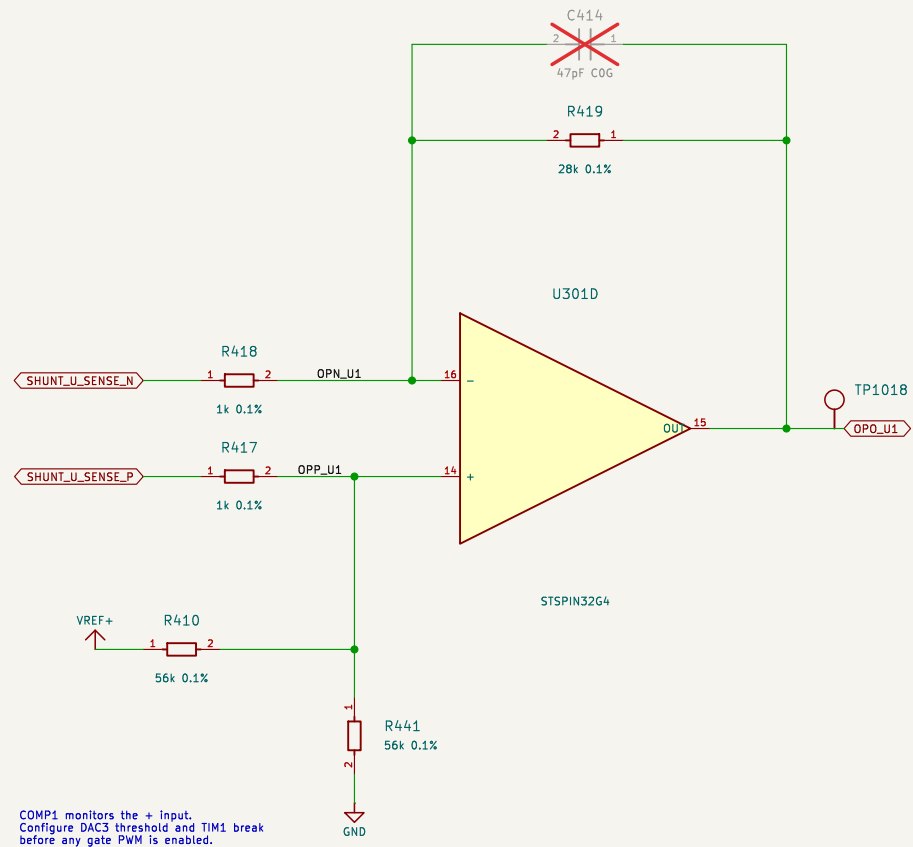
KiCad E.D.A. 9.0.4

Id: 6/12

MATCHED CURRENT CHANNELS AND LOCAL MOSFET TEMPERATURE

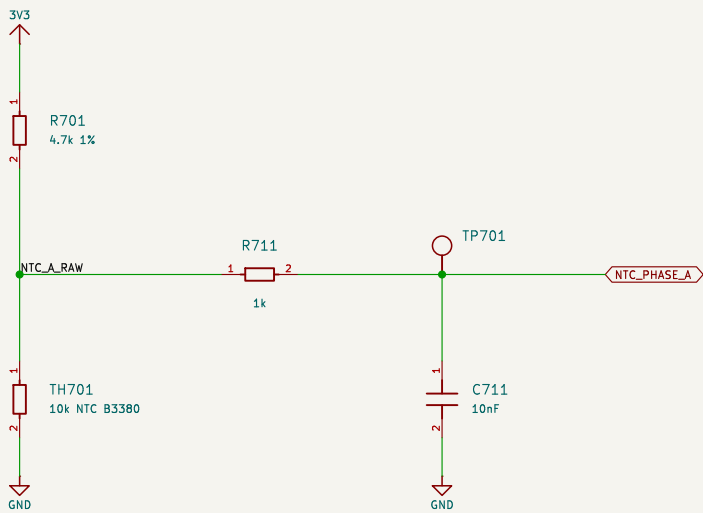
$V_{OUT} = V_{REF}/2 + 28 \times (SENSE+ - SENSE-)$. At 0.5 mOhm: 14 mV/A. Bias resistors establish the bipolar zero-current point.

PHASE U / OPAMP1



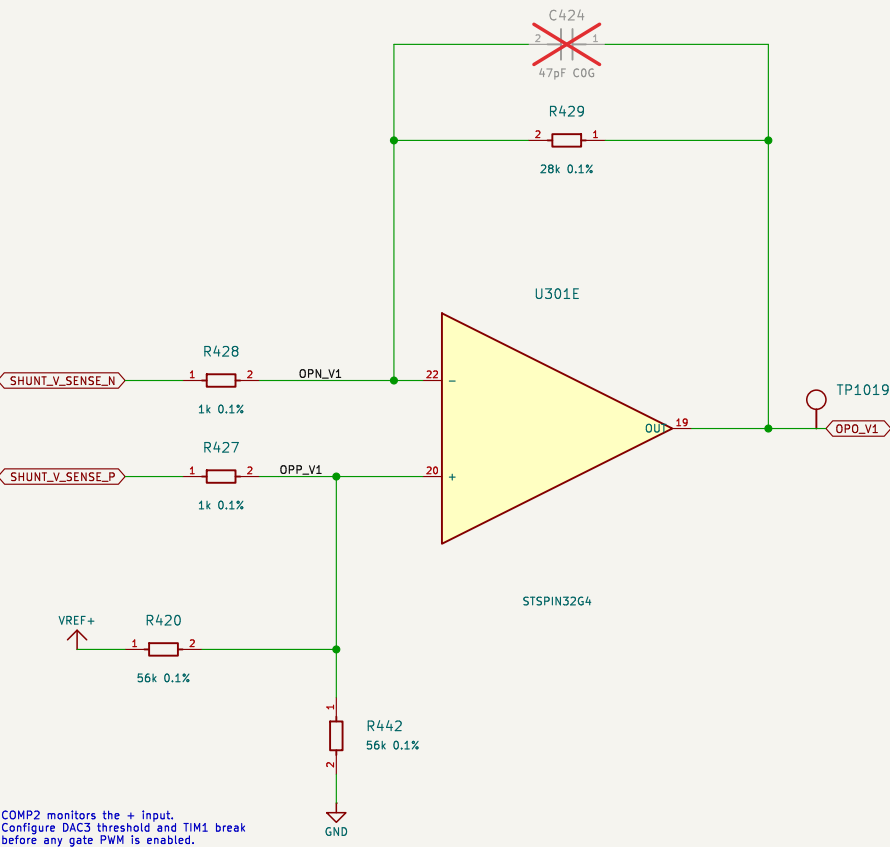
COMP1 monitors the + input.
Configure DAC3 threshold and TIM1 break
before any gate PWM is enabled.

NTC A – place beside phase U



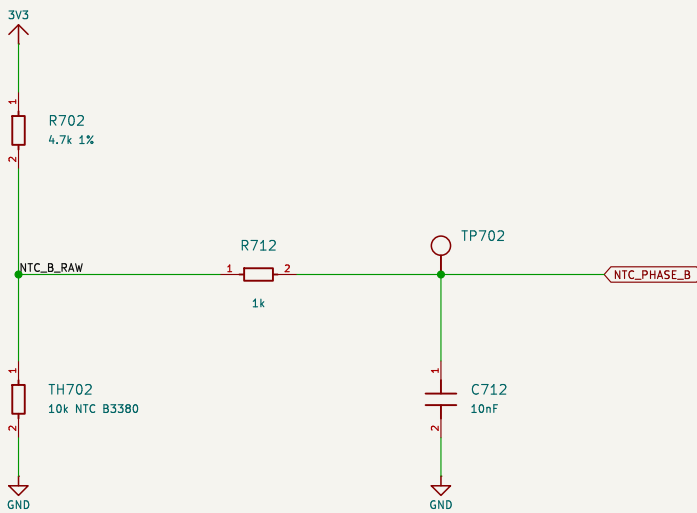
NTC faults: open -> ADC near 3V3; short -> ADC near GND. Use the hottest valid channel and calibrate protection against measured temperature.

PHASE V / OPAMP2

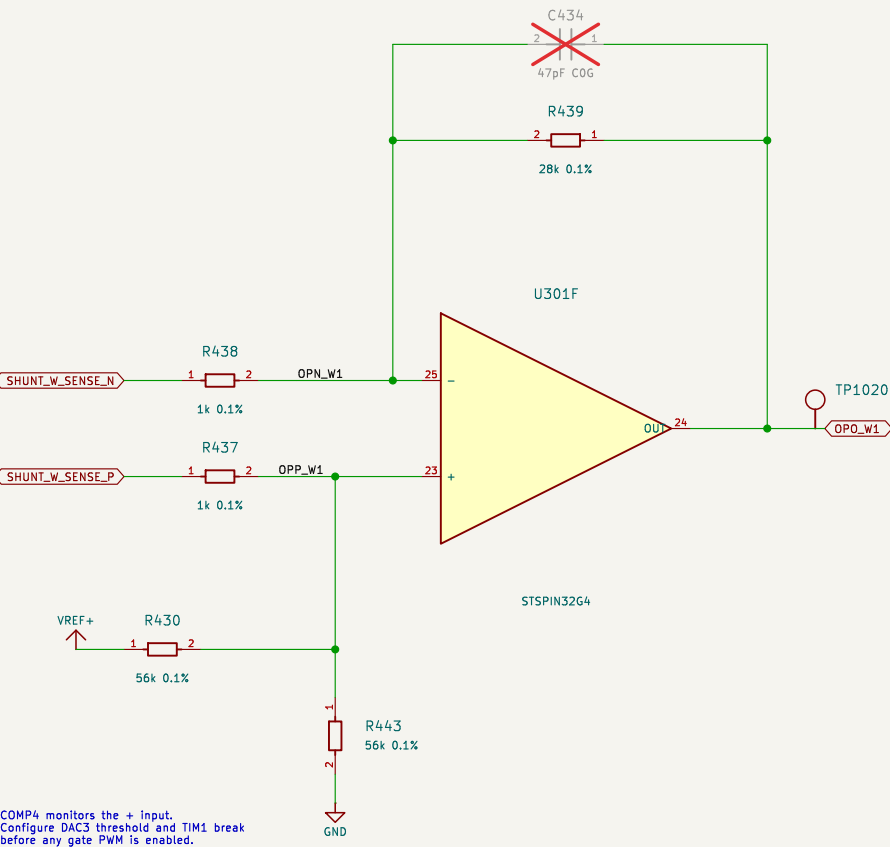


COMP2 monitors the + input.
Configure DAC3 threshold and TIM1 break
before any gate PWM is enabled.

NTC B – place beside phase V

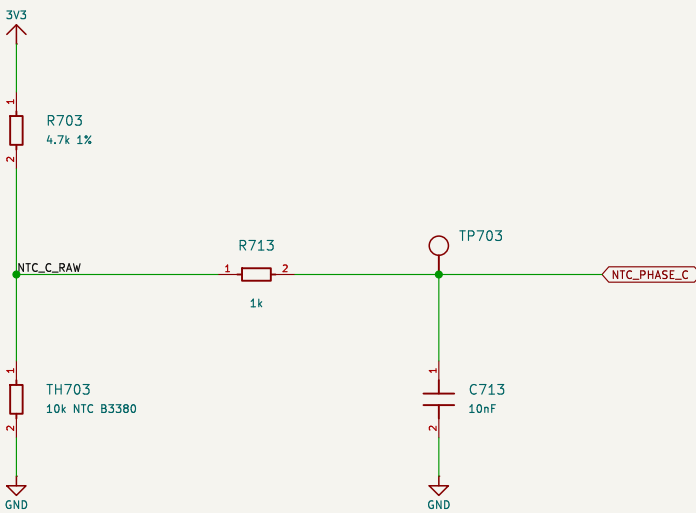


PHASE W / OPAMP3

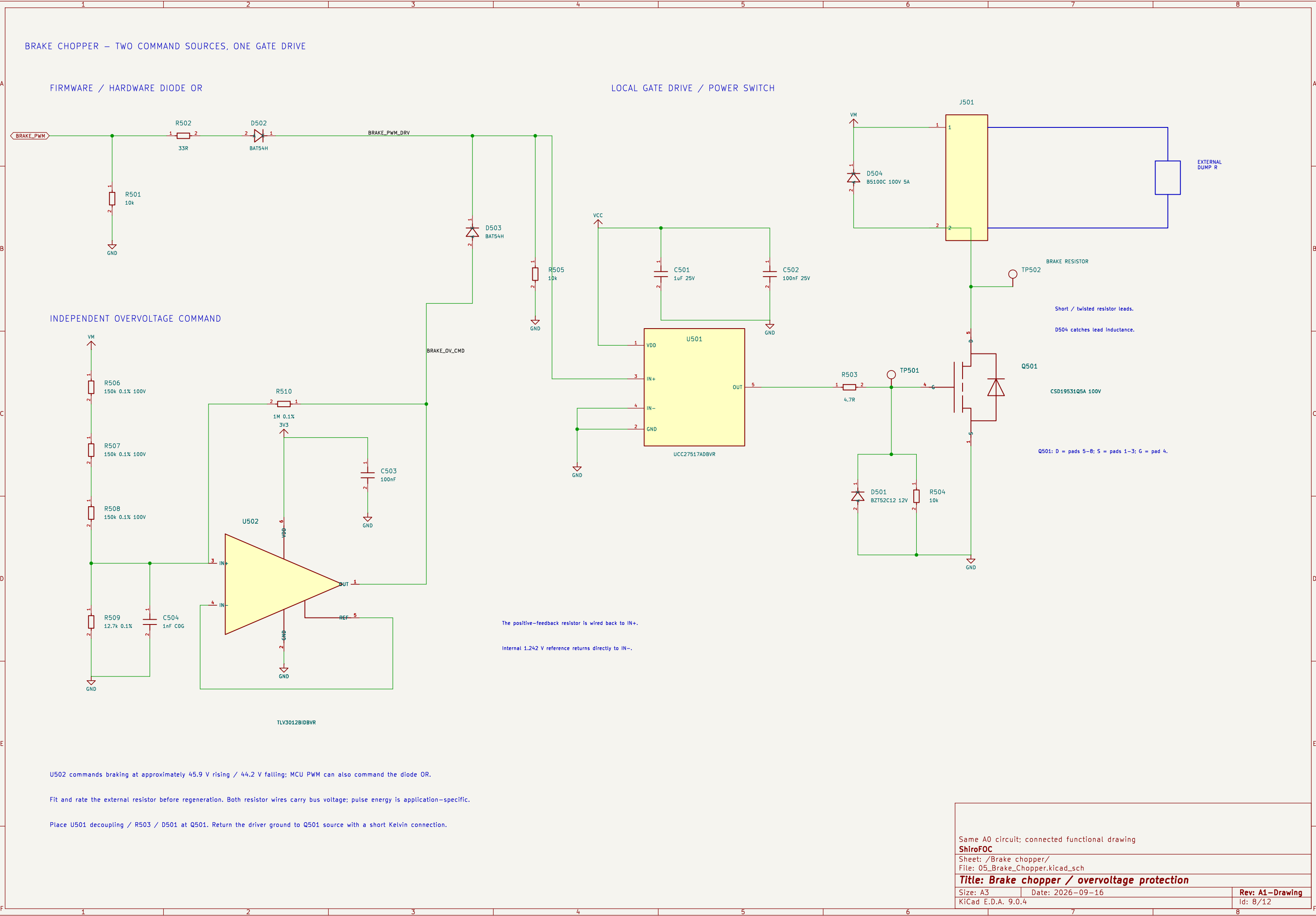


COMP4 monitors the + input.
Configure DAC3 threshold and TIM1 break
before any gate PWM is enabled.

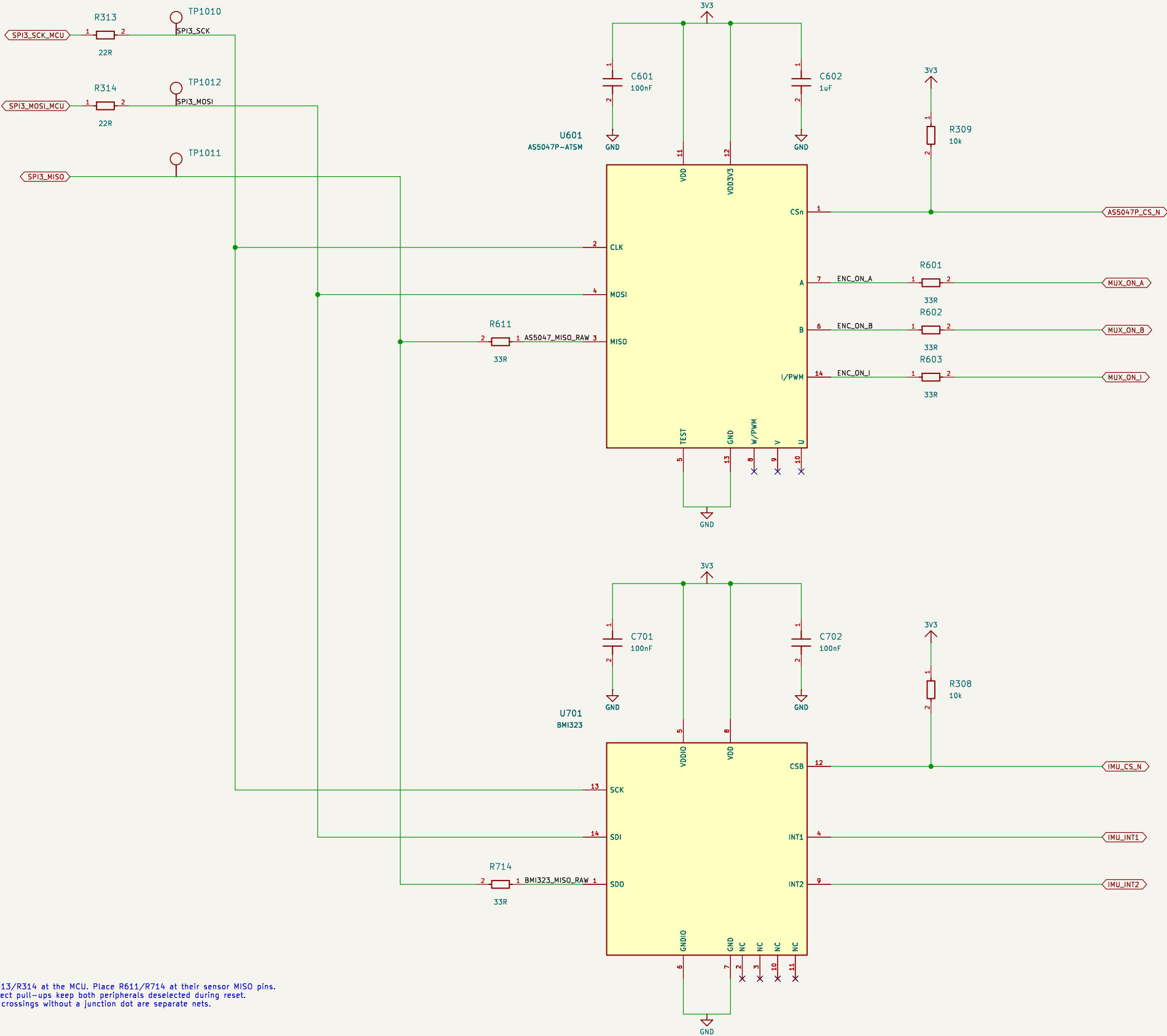
NTC C – place beside phase W



Same A0 circuit; connected functional drawing		
ShiroFOC		
Sheet: /Current / temperature/ File: 05_Current_Temperature.kicad_sch		
Title: Three Kelvin current amplifiers / phase temperature		
Size: A3	Date: 2026-09-16	Rev: A1-Drawing
KiCad E.D.A. 9.0.4		Id: 7/12



SPI3: one shared bus, separate chip selects, local return damping



ONBOARD POSITION
ABI outputs feed the selector sheet.
TEST is tied low for normal operation.
Align the sensor with the shaft magnet.

IMU
Mark X/Y axes on the PCB.
Keep power-stage heat and switching
loops away from the sensor.

Place R313/R314 at the MCU. Place R611/R714 at their sensor MISO pins.
Chip-select pull-ups keep both peripherals deselected during reset.
SPI wire crossings without a junction dot are separate nets.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /SPI sensors/
File: 06_SPI_Sensors.kicad_sch

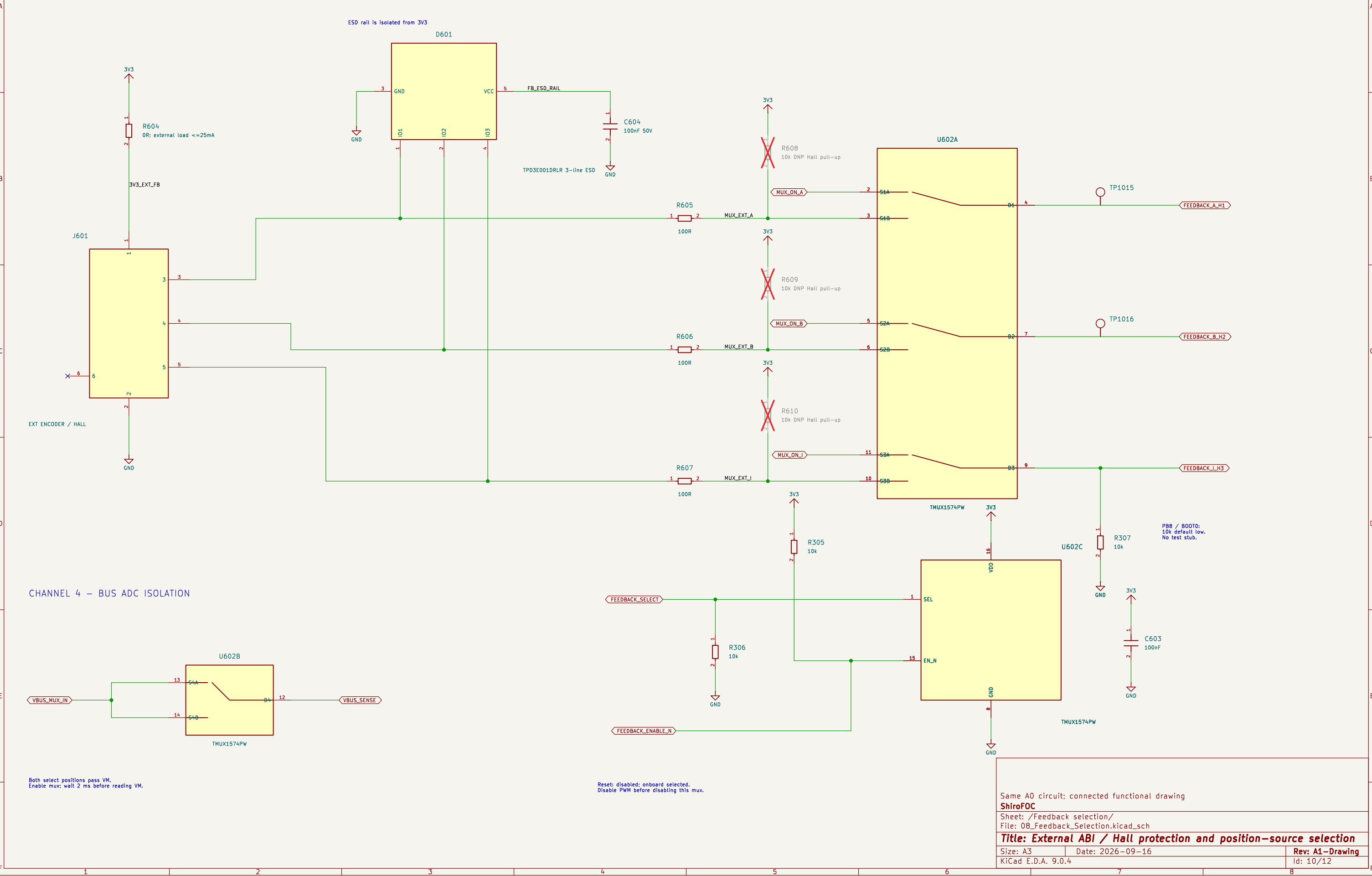
Title: Shared SPI bus / position encoder / IMU

Size: A3
Date: 2026-09-16

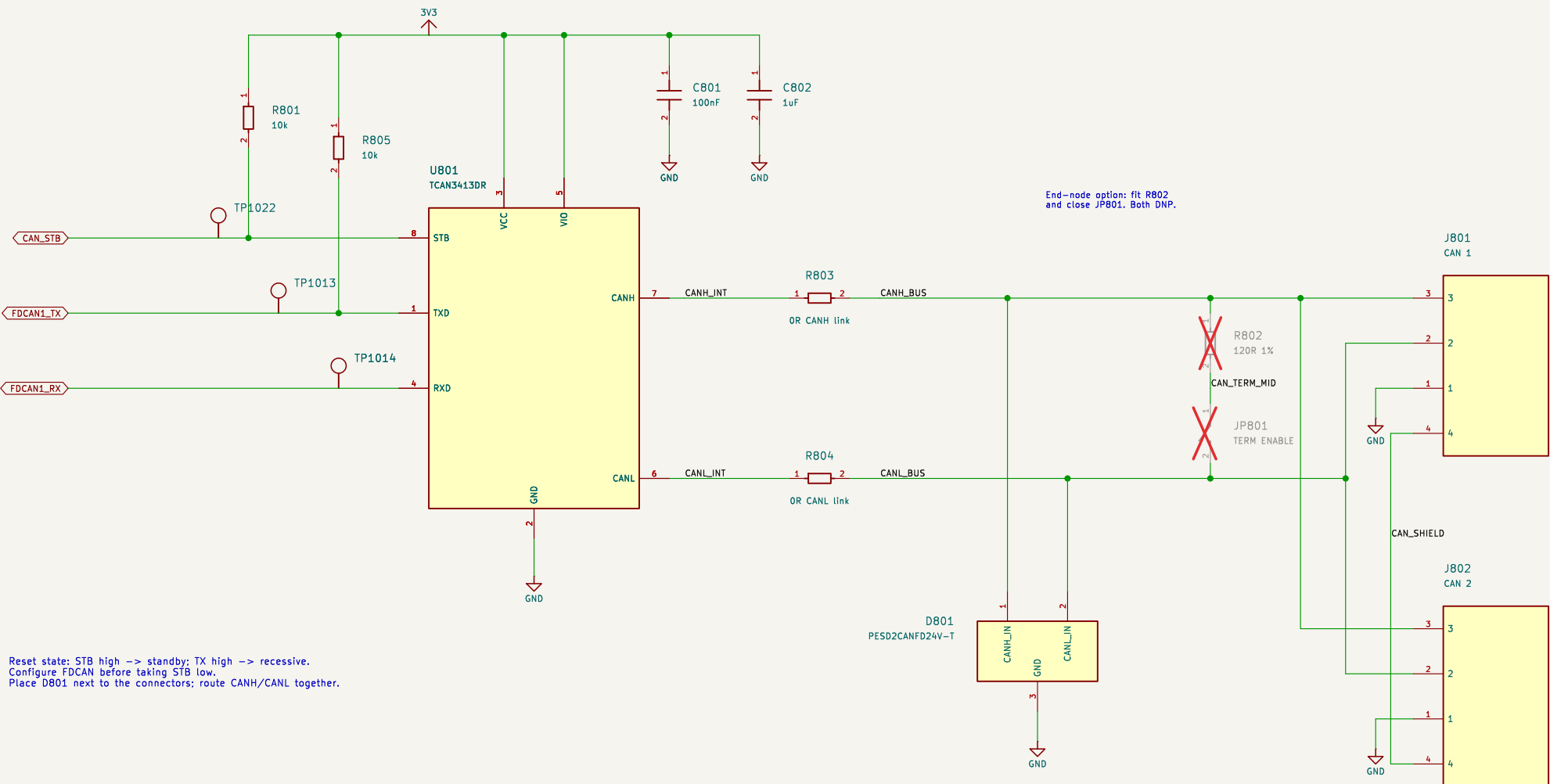
Rev: A1-Drawing
Id: 9/12

ONBOARD OR EXTERNAL POSITION FEEDBACK -> TIM4

External signals must stay within 0–3.3 V, including when board power is off. DNP pull-ups are for open–drain Hall outputs.



CAN: controller -> transceiver -> protected daisy-chain bus



Both headers: 1 GND | 2 CANL | 3 CANH | 4 shield/drain. Pin 4 carries no supply.

Same A0 circuit; connected functional drawing
ShiroFOC

Sheet: /CAN interface/
File: 08_CAN.kicad_sch

Title: CAN / CAN-FD interface

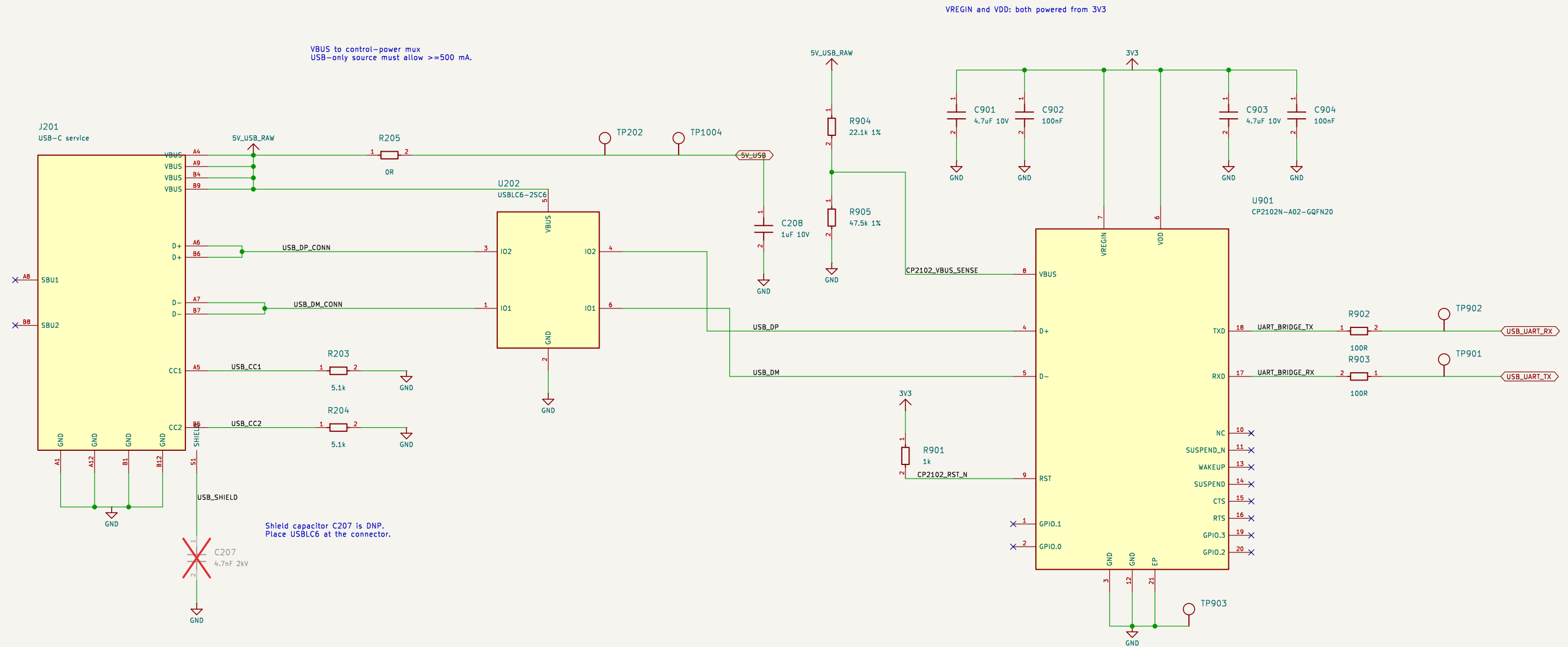
Size: A4	Date: 2026-09-16
----------	------------------

KiCad E.D.A. 9.0.4

Rev: A1-Drawing

Id: 11/12

USB service: connector, protection and bridge in one circuit



LAYOUT: route the USB pair continuously from J201 through U202 to U901; avoid stubs.
C901/C902 belong at VREGIN; C903/C904 belong at VDD. Both pin groups use the same 3V3 rail.

UART labels are named from the MCU point of view.
TXD from U901 drives MCU RX. RTS/CTS are unused.
BOOT0 and reset remain manual controls on the MCU sheet.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /USB service/

File: 09_USB_Service.kicad_sch

Title: USB-C service / UART bridge

Size: A3

Date: 2026-09-16

Rev: A1-Drawing

KiCad E.D.A. 9.0.4

Id: 12/12